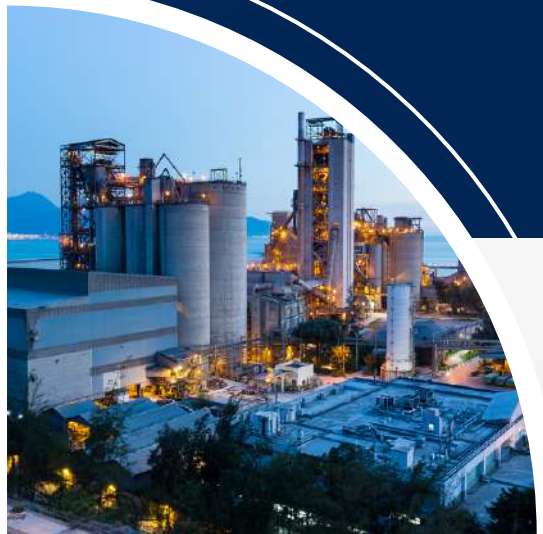


Case Study on _____

Factory Automation System



Project Overview

This project is a comprehensive solution designed to **streamline production management** and **visualization** in **manufacturing systems**. It focuses on **mapping** the journey of **raw materials** as they are processed by **machines**, resulting in **finished products** and **waste materials**.

The project features a **graph-like visualization** to illustrate **material flow**, machine interactions, and outputs, providing insights into **production efficiency**. Additionally, it integrates an employee attendance system to monitor **workforce participation** and **productivity**.

It also includes **Gantt charts** to effectively **plan, schedule, and track production timelines**, ensuring **optimized resource allocation** and **project execution**. Together, these features aim to enhance **operational transparency** and **efficiency** in **production environments**.

The screenshot displays the ASPATICS Dashboard interface. On the left, a dark blue sidebar contains a menu with the following items: Dashboard (selected), Administration, Configuration, Planning, Operations, and Analytics. The main content area is titled 'Dashboard' and shows a large empty space with a small toolbar containing a plus sign, a document icon, and a trash icon. On the right side, there is a 'PROPERTIES' panel for 'Node ID - 11'. This panel includes several input fields and dropdown menus: 'Node Type' (Please Select), 'Label' (text input), 'Node Category' (Please Select), 'Item Description' (text input), 'X Position' (text input), 'Y Position' (text input), 'Width' (text input), 'Height' (text input), 'Unit-1 Measurable' (Please Select), 'Unit-2 Measurable' (Please Select), 'Source Position' (Please Select), and 'Target Position' (Please Select). Below these fields is an 'Upload Image' section with a text prompt 'Click to upload or drag and drop' and a note '(PNG, JPEG, PNG less than 1MB)'. At the bottom of the properties panel are two buttons: a green checkmark button and a red 'X' button. The top right corner of the dashboard shows a user profile icon labeled 'admin'.

Project Feature View

1. Dashboard

- **Properties Form:** Centralized hub to manage and input essential production properties and settings for seamless operations.

2. Administration

- **Shift Management:** Define and manage work shifts for employees to optimize production schedules.
- **Employee Management:** Maintain a database of employees, track roles, and manage individual profiles.
- **Department Management:** Organize and oversee different departments involved in the production process.

3. Configuration

- **Route Configuration:** Define production routes and material flows to streamline the manufacturing process.
- **Staff Configuration:** Assign roles, manage teams, and ensure workforce alignment with production goals.
- **Material Configuration:** Maintain details about raw materials, intermediate products, and finished goods.
- **Device Configuration:** Integrate and manage machines/devices involved in the production process for efficiency improvements.

4. Planning

- **Attendance Tracking:** Monitor employee attendance for better workforce planning.
- **Staff Allocation:** Assign employees to specific machines, tasks, or production stages.
- **Job Assignment:** Allocate and schedule jobs effectively to meet production timelines.

5. Operations

- **Jobs:** Oversee production jobs in one place.

6. Analytics

- **Job Status:** Visualize and track the progress of individual jobs within the production cycle.
- **Machine-wise Report:** Analyze machine performance and usage for efficiency improvements.
- **Material Data Report:** Access detailed insights on material usage, waste, and inventory levels.
- **Material Production Report:** Evaluate production output and material transformation rates.
- **Gantt Chart:** Plan and visualize production timelines, dependencies, and schedules for better project management.

Client Details

»» Name: **Confidential**

»» Location: **India**

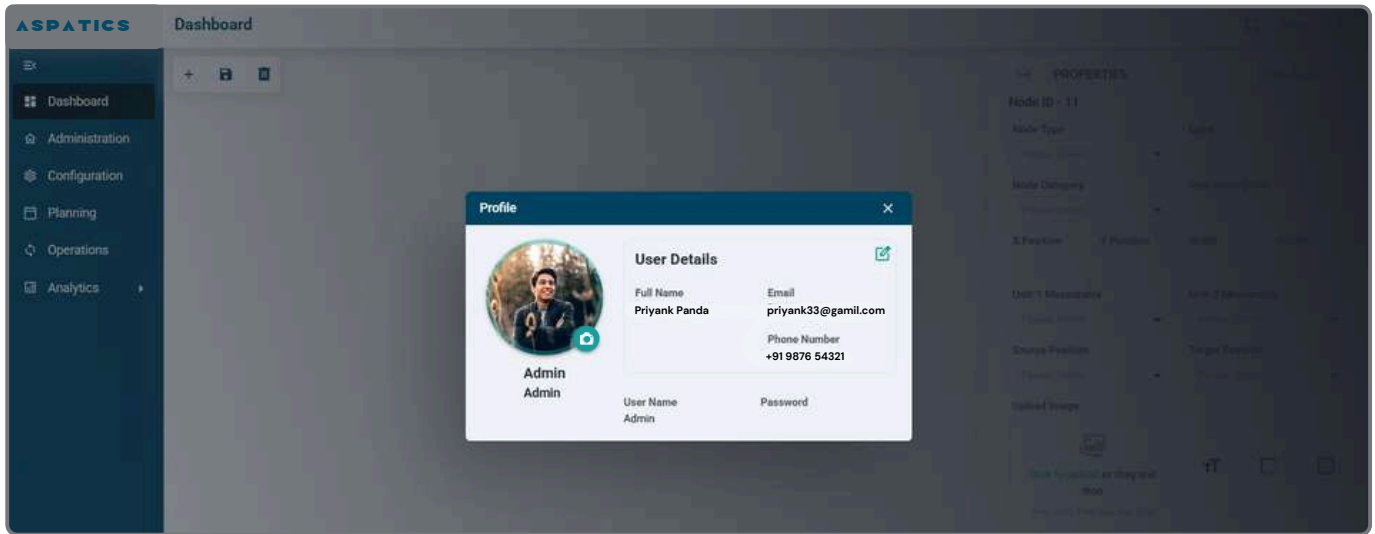
»» Industry: **Manufacturing**

Technology Details

»» **React.js**

»» **Material UI**

Screenshots



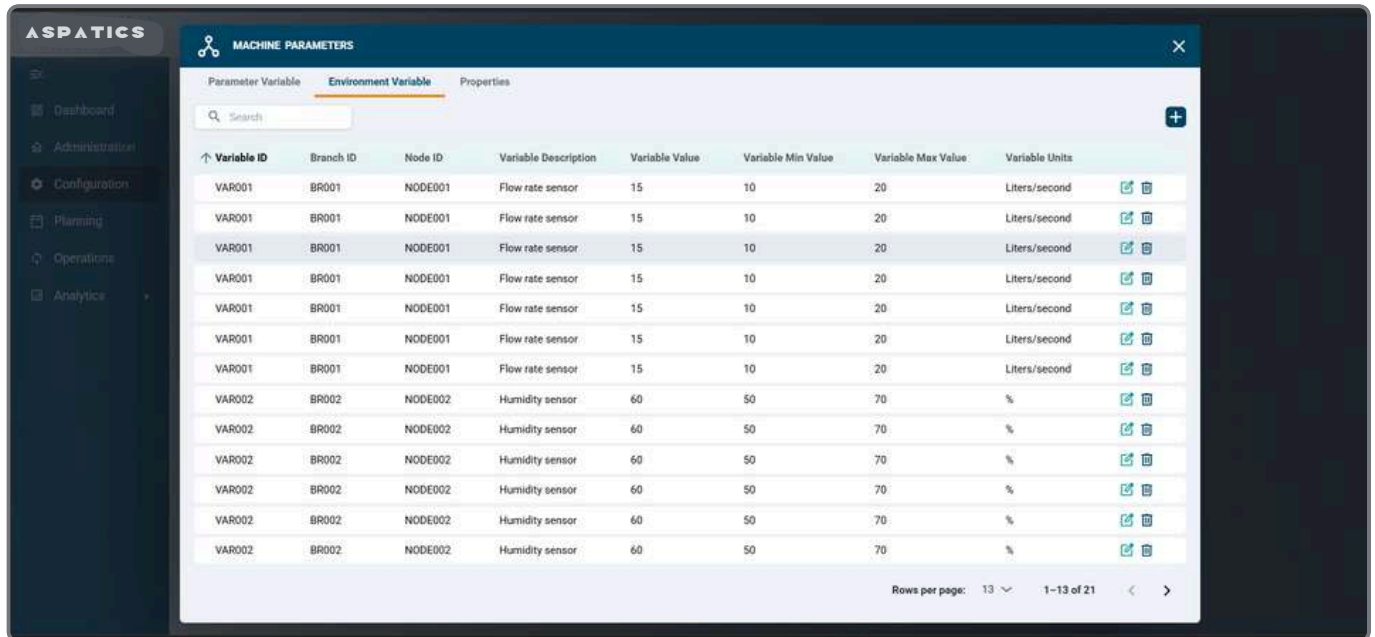
The screenshot shows the ASPATICS Dashboard. A 'Profile' modal is open, displaying user details for 'Admin Admin'. The modal includes a profile picture, a 'User Details' section with fields for Full Name, Email, and Phone Number, and a 'User Name' and 'Password' section. The background dashboard shows a sidebar with navigation options like Dashboard, Administration, Configuration, Planning, Operations, and Analytics. The main area displays a 'PROPERTIES' section for 'Node ID - 11'.

Profile

User Details

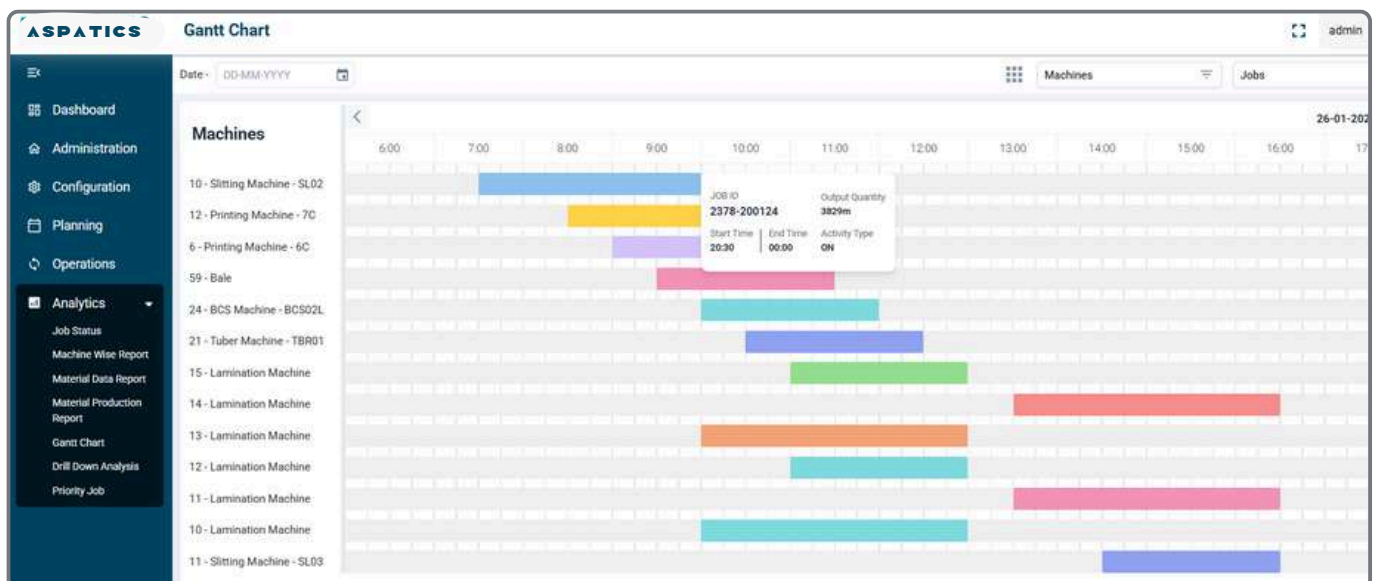
Full Name: Priyank Panda
Email: priyank33@gamil.com
Phone Number: +91 9876 54321

User Name: Admin
Password:



The screenshot shows the ASPATICS MACHINE PARAMETERS table. The table has columns for Variable ID, Branch ID, Node ID, Variable Description, Variable Value, Variable Min Value, Variable Max Value, and Variable Units. The table is filtered by 'Environment Variable' and shows 21 rows. The 'Rows per page' is set to 13, and the current page is 1-13 of 21.

Variable ID	Branch ID	Node ID	Variable Description	Variable Value	Variable Min Value	Variable Max Value	Variable Units
VAR001	BR001	NODE001	Flow rate sensor	15	10	20	Liters/second
VAR001	BR001	NODE001	Flow rate sensor	15	10	20	Liters/second
VAR001	BR001	NODE001	Flow rate sensor	15	10	20	Liters/second
VAR001	BR001	NODE001	Flow rate sensor	15	10	20	Liters/second
VAR001	BR001	NODE001	Flow rate sensor	15	10	20	Liters/second
VAR001	BR001	NODE001	Flow rate sensor	15	10	20	Liters/second
VAR001	BR001	NODE001	Flow rate sensor	15	10	20	Liters/second
VAR002	BR002	NODE002	Humidity sensor	60	50	70	%
VAR002	BR002	NODE002	Humidity sensor	60	50	70	%
VAR002	BR002	NODE002	Humidity sensor	60	50	70	%
VAR002	BR002	NODE002	Humidity sensor	60	50	70	%
VAR002	BR002	NODE002	Humidity sensor	60	50	70	%
VAR002	BR002	NODE002	Humidity sensor	60	50	70	%



Thank You

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